

ADITYA BAYU PRATAMA

Supervisor MES & IT Engineer | Smart Manufacturing & Industry 4.0

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PROFESSIONAL SUMMARY

Supervisor MES & IT Engineer with more than seven years of expertise in Manufacturing Execution Systems, Industrial IoT, machine vision and smart factory automation across high-precision electronics and the oil & gas sector. Proven record of scaling an IT department from a solo role into a five-engineer team, architecting custom C#/.NET enterprise solutions governing 500+ product models, and delivering Industry 4.0 digital transformations end to end. Two-time 1st place winner of the Giken Global Value Engineering Award (2023, 2025). Bachelor of Applied Science in Mechatronics, Robotics & Automation Engineering, Cum Laude (GPA 3.81/4.00).

CORE COMPETENCIES

- Software & Languages:** C# (.NET Framework), SQL/MySQL, Python, React.js, JavaScript, C++, LabVIEW, Dart/Flutter, R, REST APIs
- Smart Manufacturing:** MES architecture, Industrial IoT (IIoT), traceability systems, Pokayoke, AOI machine vision, SMT process, thermal print automation, Node-RED
- Hardware & Automation:** PLC (BNSP certified), SCADA, Raspberry Pi, Arduino, Lidar and laser sensors, data acquisition (DAQ), PID control
- AI & Modern Tooling:** AI-assisted development, prompt engineering, LLM integration, LM Studio (local models), Claude Code, Cursor, Codex, OpenCV/computer vision, machine learning practice
- Engineering Quality:** FMEA, DMAIC, Six Sigma practices, 5M change management, value engineering, BOM & QCPC, new product introduction
- IT & Leadership:** Team recruitment and management, department budget ownership, network infrastructure, enterprise server maintenance, DevOps basics, incident response, project management

PROFESSIONAL EXPERIENCE

- GIKEN PRECISION INDONESIA — BATAM, INDONESIA** Apr 2019 – Present
- Supervisor MES & IT Engineer** Jul 2026 – Present
- Lead the MES and IT engineering function: system roadmap, architecture standards, project prioritisation across production and support, department budget, and development of the engineering team.
 - Own the Giken One platform strategy covering MES, traceability, feeder asset control and the internal web applications the plant depends on.
 - Report department performance to management: uptime, ticket resolution, project progress and cost savings from digitalisation.
- Lead Software & IT Engineer** Apr 2019 – Jun 2026
- Designed and deployed a custom Windows-based Manufacturing Execution System (C#/.NET, MySQL) governing 500+ product models across automotive, medical and consumer electronics lines, enforcing Pokayoke defect prevention and CTQ quality compliance.
 - Built Giken One, an enterprise web platform (React.js, MySQL) digitising employee records, org charts, overtime approvals, IT ticketing and project scheduling.
 - Engineered an automated ESD wristband monitoring gate (Arduino, C#, MySQL) enforcing 100% compliance before line entry and tracking wristband defect rates.
 - Automated QR-code thermal label printing across new electronic lines through a direct C# and SQL printer interface, eliminating printing delays.
 - Established centralised network file servers across all departments, removing the risk of data loss from single-drive failure.
 - Recruited, mentored and scaled the software and IT engineering department from a solo role into a five-engineer team.
 - Contained a worm malware outbreak across 20 laptops within 3 hours with no production downtime; sustained 99.9% uptime on core MySQL servers; resolved factory network outages inside 15-minute SLAs.
- Software Test Engineer** Apr 2019 – Apr 2024
- Developed machine testing software for new electronic products and ran production trials before mass production.
 - Translated customer requirements into test criteria and validated measurement accuracy between test equipment, PLCs and the MES database.
- New Product Introduction Engineer** Apr 2019 – Sep 2021
- Studied new electronics projects from customer drawings to line readiness.

- Created and analysed FMEA, 5M change management, BOM lists and QCPC documents; ran trials and solved production problems with quality and process engineering.

CLADTEK BI-METAL MANUFACTURING — BATAM, INDONESIA

Apr 2018 – Jan 2019

Junior Programmer — R&D Department

Apr 2018 – Jan 2019

- Built an autonomous pipe-inspection robot (C#, Raspberry Pi, Lidar, light sensors) navigating inside industrial pipes, streaming live video and calculating weld defect location and severity.
- Developed low-latency data transmission protocols connecting embedded sensors to central servers for real-time welding QA evaluation.

SELECTED PROJECTS

Manufacturing Traceability System — Giken One MES (2019–present) — Plant-wide MES with per-line dashboards, full unit traceback and SMT refill control across 500+ product models. 1st place globally, Giken Value Engineering Award 2023.

Intelligent Machine Data (2025) — Real-time machine data acquisition and analytics replacing manual reporting with live line-efficiency and downtime visibility. 1st place globally, Giken VE Award 2025.

Auto Feeder Machine Data Generator (2026) — Digitalised the SMT feeder data flow across four roles: 5–7 hours with 4–6 people became a 30-minute automated flow, cutting cycle time by roughly 93%.

PCBA Management Apps (2026) — React.js and MySQL platform running the PCBA IT department: tickets, tasks, schedules, projects, attendance, assets and knowledge base on one dashboard.

SMT Pick-Up Material Analytics Dashboard (2026) — Automated analysis of component pick-up data across 23 SMT lines: 7.9 million pickups, error rate and spoilage PPM per line and machine, plus high-risk feeder detection.

Digital Feeder Inventory & Tracking (2026) — Real-time feeder location and asset control linking the pick-up material database to the refill system, tracking every feeder to area, line, machine, table and feeder number.

Multiple Remote Desktop Protocol / Centralisation Data Management (2020–2021) — Remote access architecture reaching one Windows host from Windows, Linux, Raspberry Pi, Android and iOS; centralised plant storage on servers and network shares.

LabVIEW AOI PCB Inspection System (2021) — Automated Optical Inspection algorithm in LabVIEW Vision detecting missing or misaligned components on PCB assembly lines, reducing manual QA headcount.

AI-Driven Grid Trading Application (2026) — High-frequency grid trading bot in React.js with the Telegram Bot API, using zero-reasoning AI models for low-latency execution on market sentiment and Binance API data.

IoT Emergency Response Robot & Multi-Biometric Access Controller (2020–2021) — Raspberry Pi rescue robot with thermal, gas, smoke and PIR sensors on a Node-RED dashboard; door lock with Python/OpenCV face recognition, fingerprint, RFID and PIN.

HONORS & AWARDS

1st Place — Giken Global Value Engineering Award (2025) — Top global prize for Intelligent Machine Data, optimising real-time analytics and production line efficiency.

1st Place — Giken Global Value Engineering Award (2023) — Won on first entry with the Manufacturing Traceability System, eliminating defect passage between processes.

EDUCATION

STATE POLYTECHNIC OF BATAM

Jul 2017 – Jul 2021

Bachelor of Applied Science (S.Tr.T.), Mechatronics, Robotics & Automation Engineering — Graduated Cum Laude, GPA 3.81/4.00

Coursework: machine vision (C# & LabVIEW), distributed control systems, SCADA, PLC, industrial IoT, microcontrollers, artificial intelligence and robotics, pneumatic and hydraulic automation, PID control.

LICENSES & CERTIFICATIONS

Google Data Analytics Professional Certificate (Coursera, 2022) · PLC Competency, BNSP Indonesia (2019) · Electrical Power Installation Technician, LPJK (2020) · DevOps, Computer Networks, Dart and Flutter (Dicoding, 2023)